Y10/MA/B · MATHEMATICS · PERIOD 2

Adding & subtracting *algebraic fractions*

$$\frac{3}{x-1}+\frac{2}{x+1}=\frac{5x+1}{(x-1)(x+1)}$$

Student Book 1 · 2nd edition Algebra 6 · **#88** pp. 431–433 · Examples 3–6 · Exercises 135, 136

RETRIEVE & CONNECT 0–5 MIN

Do Now — *four on the whiteboard*

1

$$\frac{1}{2} + \frac{1}{3}$$

NUMBERS

$$\frac{5}{6}$$

2

$$\frac{3}{4} - \frac{1}{6}$$

NUMBERS

$$\frac{7}{12}$$

3

$$\frac{3x+3}{2} \times \frac{x+1}{3}$$

LAST LESSON · EX 137 Q1

$$\frac{(x+1)^2}{2}$$

4

$$\frac{x}{3} + \frac{x}{6}$$

LETTERS, NUMBERS UNDERNEATH

$$\frac{x}{2}$$

Question 3 needed no new denominator. Questions 1, 2 and 4 did.
What did you have to do before you could add or subtract?

FINISHED? $\frac{x+4}{5} + \frac{x}{3}$

RETRIEVE & CONNECT 5–7 MIN

Where we are going

All add and subtract algebraic fractions with **numbers** in the denominator, e.g. $\frac{x}{3} + \frac{x+1}{2}$

Exercise 135

Most ...with **x** in the denominator, e.g. $\frac{3}{x-1} + \frac{2}{x+1}$, leaving the denominator factorised

Exercise 136 · Examples 4–5

Some factorise a denominator to find the lowest common denominator, cancel at the end, and explain **why** the answer is right

Example 6 · Exercise 136 and 136*

I CAN...

1. find the lowest common denominator (LCD)
2. write each fraction over the LCD
3. keep the brackets on the top until the fractions are joined
4. simplify the numerator, then look for a factor to cancel
5. leave the denominator factorised

EXAM FOCUS 5–7 MIN

How the exam asks it

4MA1 1H · 2025 JUNE · Q12

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

(3)

(Total for Question 12 is 3 marks)

Paper 1H, **every year 2018–2025** · about **3½ minutes** (1.2 min per mark) · usually 3 marks, sometimes combined with × ÷ for 4–5 marks.

M1 two correct fractions with a **common denominator**, or a single correct fraction
e.g. $\frac{5(6x)}{24x} + \frac{4(x-3)}{24x}$ or $\frac{15x}{12x} + \frac{2(x-3)}{12x}$

M1 correct fraction(s) with the **brackets expanded** correctly
e.g. $\frac{15x + 2x - 6}{12x}$

A1 **simplified** answer $\frac{17x-6}{12x}$

Do not ISW incorrect simplification. Writing $\frac{17x-6}{12x} = \frac{11}{12}$ turns M1M1A1 into **M2 A0** — cancelling terms loses the mark.

RETRIEVE & CONNECT 7–9 MIN

Six words for today

denominator 分母	lowest common denominator 最小公分母
equivalent fraction 等值分式	single fraction 单一分式
expand 展开	simplest form 最简形式

Click a card to flip it · say the word in a full sentence: “To add these fractions I first need a common ____.”

EXPLAIN & MODEL 9–15 MIN

The method does not change when numbers become *letters*

WITH NUMBERS

$$\frac{1}{2}+\frac{1}{3}=\frac{3}{6}+\frac{2}{6}=\frac{5}{6}$$

$$\frac{3}{4}-\frac{1}{6}=\frac{9}{12}-\frac{2}{12}=\frac{7}{12}$$

WITH LETTERS

$$\frac{1}{x}+\frac{1}{y}=\frac{y}{xy}+\frac{x}{xy}=\frac{x+y}{xy}$$

$$\frac{3}{x-1}+\frac{2}{x+1}=?$$

Same denominator first — then add the tops.

EXPLAIN & MODEL 9–15 MIN

Numbers underneath *first*

EXAMPLE 3 · P.431 Express $\frac{3(4x-1)}{2} - \frac{2(5x+3)}{3}$ as a single fraction.

$$\frac{3(4x-1)}{2} - \frac{2(5x+3)}{3}$$

LCD

LCM of 2 and 3 is 6

$$= \frac{3 \times 3(4x-1)}{6} - \frac{2 \times 2(5x+3)}{6} = \frac{9(4x-1) - 4(5x+3)}{6}$$

$$= \frac{36x - 9 - 20x - 12}{6}$$

$$= \frac{16x - 21}{6}$$

What is the **lowest common multiple** of 2 and 3? So the LCD is 6.

Rewrite each fraction over 6: the first needs **×3** top and bottom, the second **×2**.

Note: the brackets on top are not multiplied out until the second step. This avoids sign mistakes when subtracting.

Expand, then simplify the numerator: 36x − 20x = 16x and −9 − 12 = −21.

EXPLAIN & MODEL 9–15 MIN

The minus sign hits *every* term

× THE CLASSIC SLIP

$$\frac{x+1}{3} - \frac{x-1}{2}$$

$$= \frac{2(x+1) - 3(x-1)}{6}$$

$$= \frac{2x + 2 - 3x - 3}{6}$$

$$= \frac{-x - 1}{6}$$

✓ BRACKET, THEN EXPAND

$$\frac{x+1}{3} - \frac{x-1}{2}$$

$$= \frac{2(x+1) - 3(x-1)}{6}$$

$$= \frac{2x + 2 - 3x + 3}{6}$$

$$= \frac{5 - x}{6}$$

Test it with a number: put $x = 1$. Left: $\frac{2}{3} - 0 = \frac{2}{3}$. Right (good): $\frac{5-1}{6} = \frac{2}{3}$ ✓. Right (slip): $\frac{-1-1}{6} = -\frac{1}{3}$ ×.

EXPLAIN & MODEL 9–15 MIN

Now with x in the denominator

EXAMPLE 4 · P.432 Express $\dfrac{3}{x-1}+\dfrac{2}{x+1}$ as a single fraction.

LCD $(x-1)(x+1)$

$$= \frac{3(x+1)}{(x-1)(x+1)}+\frac{2(x-1)}{(x+1)(x-1)}$$

$$= \frac{3(x+1)+2(x-1)}{(x-1)(x+1)}$$

$$= \frac{3x+3+2x-2}{(x-1)(x+1)}$$

$$= \frac{5x+1}{(x-1)(x+1)}$$

Nothing common, so the LCD is the **two brackets multiplied**.

Each fraction is missing the **other** bracket. Multiply top and bottom by it.

Add the fractions — one denominator now.

Note: expand the numerator only. Leave the denominator factorised — it is easier to simplify later.

Check with $x=3$: left $\frac{3}{2}+\frac{2}{4}=2$; right $\frac{16}{2\times 4}=2$ ✓

EXPLAIN & MODEL 15–21 MIN

Your turn at the *board*

EXAMPLE 5 · P.432 STUDENT AT THE BOARD $\dfrac{x+1}{x+2} - \dfrac{x-2}{x-1}$

LCD $(x+2)(x-1)$

$$= \frac{(x+1)(x-1) - (x-2)(x+2)}{(x+2)(x-1)}$$

$$= \frac{(x^2-1) - (x^2-4)}{(x+2)(x-1)}$$

$$= \frac{x^2 - 1 - x^2 + 4}{(x+2)(x-1)}$$

$$= \frac{3}{(x+2)(x-1)}$$

Who can tell me the LCD **without** looking?

Keep the second numerator **in a bracket**: the minus is about to hit both terms.

Expand each product in its own brackets first.

Remove the bracket: $-(x^2-4) = -x^2 + 4$

The x^2 terms cancel — the numerator is just a number. Check with $x = 2$: left $\frac{3}{4} - 0$, right $\frac{3}{4 \times 1}$ ✓

EXPLAIN & MODEL 15–21 MIN

When the denominator *factorises*

EXAMPLE 6 · P.432 B/C PREVIEW $\dfrac{2}{x-2} - \dfrac{6}{x^2-x-2}$

1 $$x^2 - x - 2 = (x-2)(x+1)$$

= $$\frac{2(x+1)}{(x-2)(x+1)} - \frac{6}{(x-2)(x+1)} = \frac{2(x+1)-6}{(x-2)(x+1)}$$

= $$\frac{2x-4}{(x-2)(x+1)} = \frac{2(x-2)}{(x-2)(x+1)}$$

= $$\frac{2}{x+1}$$

Factorise the denominator first. Now the LCD is just $(x-2)(x+1)$.

Only the first fraction needs multiplying — by the bracket it is missing.

Factorise the numerator — a factor $(x-2)$ appears!

Cancel the **factor** — not a term. This is last lesson's skill, finishing the job.

APPLY & PRACTISE 21–34 MIN

Choose your route · *one finish line*

Build — ROUTE A

Numbers in the denominator · teacher-supported start

Ex135 · 1	$\frac{x}{3}+\frac{x+1}{2}$	$=\frac{5x+3}{6}$
Ex135 · 2	$\frac{x}{2}-\frac{x+2}{4}$	$=\frac{x-2}{4}$
Ex135 · 4	$x+\frac{x+3}{4}$	$=\frac{5x+3}{4}$
Ex135 · 7	$\frac{x-3}{2}-\frac{x+1}{5}$	$=\frac{3x-17}{10}$
Ex135 · 10	$\frac{2}{3}-\frac{2-3x}{6}$	$=\frac{3x+2}{6}$
Ex136 · 1	$\frac{1}{2x}+\frac{1}{3x}$	$=\frac{5}{6x}$
Ex136 · 2	$\frac{3}{4x}-\frac{1}{2x}$	$=\frac{1}{4x}$

2025 Jun 1H Q12 · 3 marks

$\frac{5}{4}+\frac{x-3}{6x}=\frac{17x-6}{12x}$

Secure — ROUTE B

Brackets · x in the denominator

Ex135 · 9	$\frac{3x-2}{6}-\frac{3x+2}{9}$	$=\frac{3x-10}{18}$
Ex135 · 11	$\frac{2x+5}{4}-\frac{2(x-3)}{3}$	$=\frac{39-2x}{12}$
Ex135 · 12	$\frac{2(2x+1)}{5}+\frac{3(x-1)}{2}$	$=\frac{23x-11}{10}$
Ex136 · 3	$\frac{1}{2}-\frac{1}{x-2}$	$=\frac{x-4}{2(x-2)}$
Ex136 · 4	$\frac{1}{x+1}+\frac{1}{x-1}$	$=\frac{2x}{(x-1)(x+1)}$
Ex136 · 5	$\frac{3}{x-1}-\frac{2}{x+2}$	$=\frac{x+8}{(x-1)(x+2)}$

2021 Jan 1H Q12a · 3 marks

$\frac{4}{x-2}-\frac{3}{x+1}=\frac{x+10}{(x-2)(x+1)}$

Stretch — ROUTE C

Factorise the denominator · combine with ÷

Ex136 · 6	$\frac{4}{x^2}-\frac{2}{xy}$	$=\frac{4y-2x}{x^2y}$
Ex136 · 7	$\frac{1}{x}+\frac{x}{x-3}$	$=\frac{x^2+x-3}{x(x-3)}$
Ex136 · 8	$\frac{2}{x+4}+\frac{10}{x^2+3x-4}$	$=\frac{2}{x-1}$
Ex136 · 9	$\frac{3}{x+2}-\frac{3}{x^2+5x+6}$	$=\frac{3}{x+3}$
Ex136* · 5	$\frac{x+3}{x+2}-\frac{x-3}{x-2}$	$=\frac{2x}{(x+2)(x-2)}$

2017 SP 2H Q21 · 4 marks

$\frac{5}{4x^2-25}-(2x+3)\div\frac{4x^2+16x+15}{7}=\frac{40-14x}{4x^2-25}$

Finished? Ex136* Q8 → then Ex137* Q12 (page 434)

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 1 OF 8

What is the lowest common denominator of $\frac{1}{2x}$ and $\frac{1}{3x}$?

- ▲ $5x$
- ◆ $6x$ ✓
- ● $6x^2$
- ■ x

The smallest expression both $2x$ and $3x$ divide into is $6x$. Then $\frac{1}{2x} + \frac{1}{3x} = \frac{3+2}{6x} = \frac{5}{6x}$ (Ex 136 Q1).

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 2 OF 8

$$\frac{x}{3} + \frac{x}{6} =$$

▲ $\frac{2x}{9}$

◆ $\frac{x}{18}$

● $\frac{2x}{6}$

■ $\frac{x}{2}$ ✓

$\frac{2x}{6} + \frac{x}{6} = \frac{3x}{6} = \frac{x}{2}.\ \frac{2x}{9}$ adds the denominators — never do that.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 3 OF 8

Line 1 $\frac{x+1}{3} - \frac{x-1}{2}$

Line 2 $= \frac{2(x+1)-3(x-1)}{6}$

Line 3 $= \frac{2x+2-3x-3}{6}$

Line 4 $= \frac{-x-1}{6}$

Which line contains the first mistake?

- Line 3 ✓
- Line 2
- Line 4
- There is no mistake

$-3(x-1) = -3x+3$, not $-3x-3$. Correct answer: $\frac{5-x}{6}$.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 4 OF 8

$$\frac{3}{x-1}+\frac{2}{x+1}=$$

▲ $\frac{5}{2x}$

◆ $\frac{5}{x^2-1}$

● $\frac{5x+1}{(x-1)(x+1)}$ ✓

■ $\frac{5x-1}{(x-1)(x+1)}$

$\frac{3(x+1)+2(x-1)}{(x-1)(x+1)}=\frac{5x+1}{(x-1)(x+1)}$ — Example 4.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 5 OF 8

Which is equal to $\frac{1}{x} + \frac{1}{y}$?

▲ $\frac{2}{x+y}$

◆ $\frac{x+y}{xy}$ ✓

● $\frac{1}{xy}$

■ $\frac{1}{x+y}$

$\frac{y}{xy} + \frac{x}{xy} = \frac{x+y}{xy}$. Test: $x = 1, y = 2$ gives $1 + \frac{1}{2} = \frac{3}{2}$; $\frac{2}{x+y} = \frac{2}{3}$ ×.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 6 OF 8

$$\frac{x+2}{3} - \frac{x-1}{4} =$$

▲ $\frac{x+11}{12}$ ✓

◆ $\frac{x+5}{12}$

● $\frac{7x+5}{12}$

■ $\frac{x+1}{12}$

$\frac{4(x+2)-3(x-1)}{12} = \frac{4x+8-3x+3}{12} = \frac{x+11}{12}$. $\frac{x+5}{12}$ forgets that $-3 \times -1 = +3$.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 7 OF 8

$$\frac{1}{x-2} - \frac{1}{x+2} =$$

▲ 0

◆ $-\frac{4}{x^2-4}$

● $\frac{2}{x^2-4}$

■ $\frac{4}{x^2-4}$ ✓

$\frac{(x+2)-(x-2)}{(x-2)(x+2)} = \frac{4}{x^2-4}$. The two fractions are **not** the same, so it is not 0.

REVIEW & CONSOLIDATE 34–40 MIN

QUESTION 8 OF 8

To combine $\frac{2}{x-2} - \frac{6}{x^2-x-2}$, what is the best first step?

- Add the denominators
- Multiply the two denominators
- Factorise $x^2 - x - 2$ ✓
- Cancel the 2 and the 6

$x^2 - x - 2 = (x-2)(x+1)$, so the LCD is only $(x-2)(x+1)$ — Example 6.

REVIEW & CONSOLIDATE 34–40 MIN

Team scores



0

CLICK +1 · RIGHT-CLICK −1



0

CLICK +1 · RIGHT-CLICK −1



0

CLICK +1 · RIGHT-CLICK −1



0

CLICK +1 · RIGHT-CLICK −1

REVIEW & CONSOLIDATE 40–42 MIN

Three rules to take away

1 Make the _____ the same first — use the _____ one.

$\frac{1}{2}+\frac{1}{3}$ and $\frac{3}{x-1}+\frac{2}{x+1}$ — same method.

2 Keep the _____ on top until the fractions are joined. Then _____ the numerator only.

$-3(x-1)=-3x+3$ — the minus hits every term.

3 Factorise the _____, then cancel _____ — never terms.

$\frac{2(x-2)}{(x-2)(x+1)}=\frac{2}{x+1}$ but $\frac{17x-6}{12x}\neq\frac{11}{12}$

Click the gaps to reveal the keywords — students say them first.

REVIEW & CONSOLIDATE 42–45 MIN

Before you go

EXIT TICKET · INDIVIDUAL · COLLECTED AT THE DOOR

1 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest form.

4MA1 1H · 2023 Nov · Q11 · 3 marks · M1 M1 A1

2 $\frac{5}{6} - \frac{1}{4}$

3 A student writes $\frac{1}{x} + \frac{1}{y} = \frac{2}{x+y}$. Use $x = 1,\ y = 2$ to show it is wrong.

HOMEWORK · DUE NEXT LESSON

TEXTBOOK	Exercise 135: Q3, Q5, Q6
TEXTBOOK	Exercise 136: Q6
EXAM	2019 Jan 1H Q11 · 3 marks
STRETCH	Ex 135* Q9, Q10 · Ex 136* Q2, Q8

Attempt before checking the printed answers; the official mark scheme for the exam question is on the last page of the pack.